PaddyGuard AI:

Multimodal Intelligence for Rice Disease & Pest Management



Project ID : R26-SE-015

Problem

- Late Disease Identification – Farmers rely on experience or delayed expert advice, causing wrong treatments and 30% - 80% crop losses.
- English-Only Systems – Most tools support only English, making them difficult for Sinhala-speaking farmers.
- Fragmented Solutions – No single platform integrates disease detection, pest identification, and treatment advice.

Design Excellence

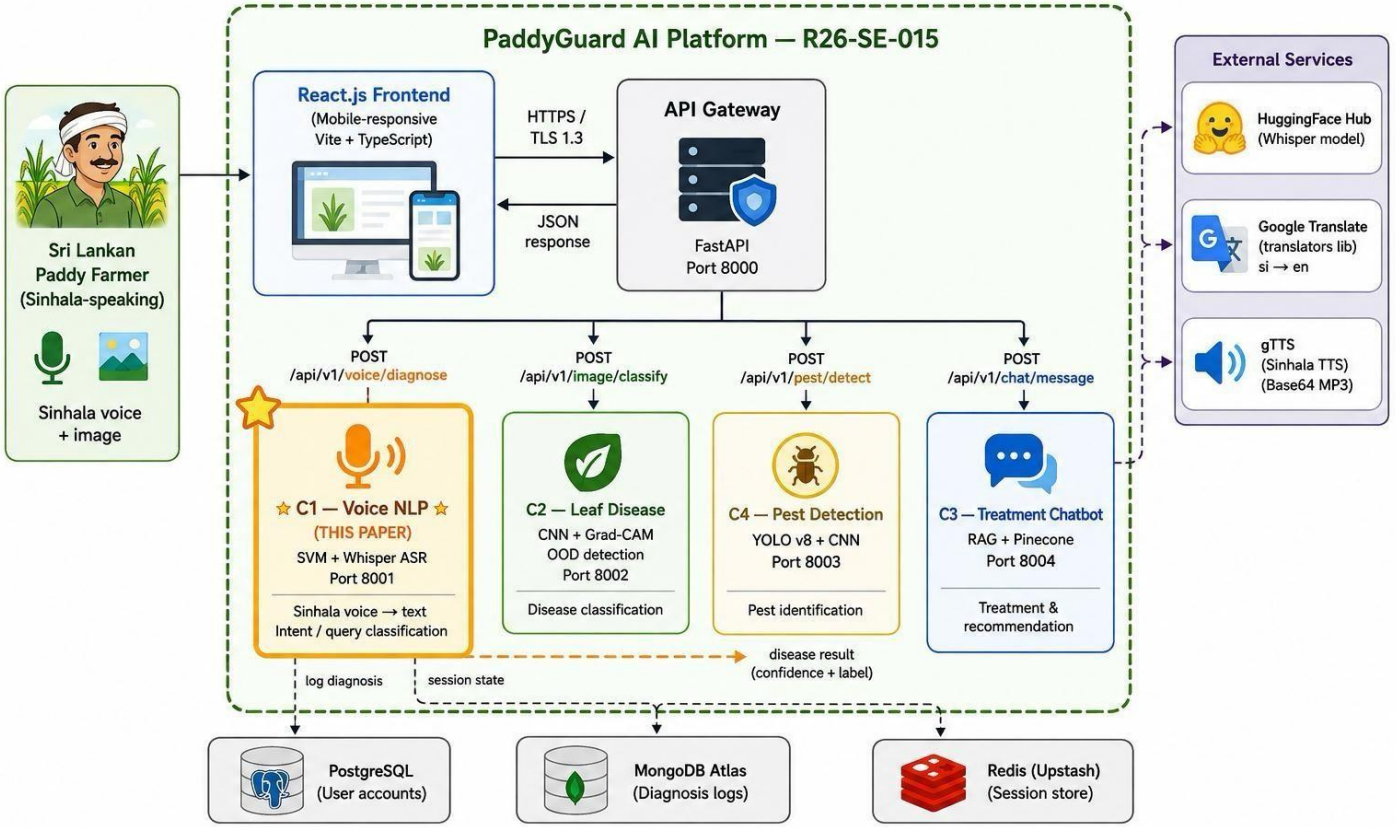
- Sinhala voice-based disease diagnosis.
- AI-based rice leaf disease detection.
- Pest identification using deep learning.
- OOD detection for unknown inputs.
- Smart treatment advice through an AI chatbot.



Comments for the PP2 and how team

STUDENT'S NAME & ID	COMMENTS FOR THE PP2
IT22273680	User testing should be done validate the accuracy of the component
it22168740	User testing should be done validate the accuracy of the component
IT22065858	User testing should be done validate the accuracy of the component
IT22303820	Just the implementation is done. need to validate the model and response accuracy. Do a user testing

PaddyGuard AI: High-Level Architecture Diagram.



Voice-Based Rice Disease Diagnosis Using Speech Recognition and NLP



IT22273680 | Weerasinghe K G J P

- **Sinhala Voice Recognition:** Used Whisper-based speech recognition to convert farmers' Sinhala voice input into text.
- **Disease Classification:** Developed an NLP-based SVM model to identify rice diseases from reported symptoms, achieving a 96.67% F1-score.
- **Smart Diagnosis:** Implemented confidence-based follow-up questions, out-of-domain input detection, and Sinhala text and voice feedback.

Rice Leaf Disease Classification with Out-of-Distribution Detection



IT22168740 | Hewanayake H.M.L.M

- **Rice Leaf Disease Classification:** Developed a CNN-based system to identify rice leaf diseases from uploaded images.
- **Out-of-Distribution (OOD) Detection:** Implemented methods to reject non-rice and unknown images instead of forcing incorrect disease predictions.
- **Visual Explainability:** Used Grad-CAM to highlight affected regions of rice leaves and help users understand model predictions.

Rice Pest Detection with Out-of-Distribution Detection



IT22065858 | E P K Kalhara

- **AI Pest Detection:** Developed a DenseNet121-based model to identify rice pests from uploaded images.
- **OOD Detection & Explainable AI:** Implemented unknown-image detection and Grad-CAM visualizations to highlight image regions influencing pest predictions.
- **System Integration:** Integrated pest prediction, confidence scores, image analysis, and detection history with the PaddyGuard AI application.

Treatment Advisory System and Agricultural Chatbot



IT22303820 | Keshan B K

- **AI-Based Treatment Recommendations:** Provides rice farmers with treatment guidance, pesticide dosage information, and preventive farming advice based on disease or pest diagnosis.
- **RAG-Based Knowledge Retrieval:** Retrieves relevant agricultural information from trusted sources such as IRRI and FAO to support reliable recommendations.
- **Multilingual Agricultural Chatbot:** Uses NLP to understand farmer queries and provides context-aware responses with Sinhala and English language support.

Results

- Achieved a 96.67% F1-score for Sinhala voice-based symptom classification using the SVM model.
- Achieved over 92% accuracy in rice leaf disease detection using CNN models.
- Implemented AI-based pest detection for five pest classes.
- Integrated Sinhala voice-based diagnosis with the treatment recommendation system.
- Developed an integrated solution for rice disease detection, pest identification, and agricultural guidance.

LIMITATIONS

- The system currently supports only selected rice diseases and five pest classes.
- Treatment recommendations are based on the information available in the knowledge base.
- Voice-based diagnosis is designed for Sinhala symptom descriptions.
- The system is intended to support farmers and does not replace professional agricultural advice.

Future Improvements

- Expand the system to support more rice disease and pest classes.
- Enhance Sinhala voice recognition for different regional accents.
- Improve treatment recommendations by expanding the agricultural knowledge base.
- Develop a mobile-friendly version for easier access by farmers.
- Integrate more real-world data to further enhance system performance.



Thank You